

Backend + Agentic AI + DevOps — Complete Learning Roadmap

PHASE 1: Strong Backend Foundations (Must-Have)

1 Programming & Core CS

- Python (advanced): async/await, generators, decorators
- Data Structures & Algorithms (for interviews + optimization)
- OOP & SOLID principles
- Design patterns (Repository, Factory, Singleton)

2 Backend Frameworks

- **FastAPI (Mastery)**
 - Dependency Injection
 - Background tasks
 - Middleware
 - WebSockets
- Django (optional but good for jobs)

3 Databases

- SQL: PostgreSQL (indexes, joins, transactions)
- ORM: SQLAlchemy / SQLAlchem
- NoSQL: MongoDB
- Redis (caching, queues)

4 Authentication & Security

- JWT & OAuth2
- Role-Based Access Control (RBAC)
- Password hashing
- Rate limiting
- CORS & CSRF

5 API Design

- REST best practices
- Pagination, filtering, versioning
- OpenAPI / Swagger
- Error handling & logging

PHASE 2: AI Fundamentals (Required for Agentic AI)

6 AI & ML Basics

- Machine Learning concepts
- NLP basics
- Vector embeddings
- Similarity search

7 LLM Foundations

- How LLMs work (tokens, context, temperature)
- Prompt engineering
- OpenAI / Gemini / Anthropic APIs
- Structured outputs (JSON schemas)

PHASE 3: Agentic AI (CORE PHASE)

8 Agent Concepts

- What is an AI Agent?
- Tool calling
- Planning vs execution
- Memory (short-term & long-term)
- Reflection & self-correction

9 OpenAI Agents SDK (You are already here)

- Agent creation
- Function tools
- Agents as tools
- Multi-agent systems
- Streaming responses

Key Skills to Master

- Intent detection
- Autonomous decision making
- Task decomposition
- Error recovery

1 0 Agent Memory & RAG

- Vector databases (Pinecone, FAISS)
- Document ingestion

- Embedding pipelines
- Hybrid search

1 **1** Real Agent Projects

- AI Todo Agent (CRUD via tools)
- Blog Generator Agent
- Research Agent
- DevOps Assistant Agent
- Browser Automation Agent (Playwright)

PHASE 4: DevOps Foundations (CRITICAL)

1 **2** Linux & Networking

- Linux commands
- File permissions
- Process management
- Nginx basics
- HTTP / HTTPS

1 **3** Git & GitHub

- Git workflows
- PR reviews
- GitHub Actions

PHASE 5: Containerization & Deployment

1 **4** Docker

- Dockerfile
- Docker Compose
- Multi-stage builds
- Container security

1 **5** CI/CD Pipelines

- GitHub Actions
- Auto testing
- Auto deployment

PHASE 6: Cloud & Infrastructure

1 Cloud Platforms

- AWS (EC2, S3, RDS, IAM)
- GCP or Azure (basic)

1 Infrastructure as Code

- Terraform basics
- Environment management

1 Kubernetes (Advanced)

- Pods, Services
- ConfigMaps & Secrets
- Auto-scaling

PHASE 7: Monitoring & Scaling

- Logging (ELK stack)
- Monitoring (Prometheus, Grafana)
- Load balancing
- Horizontal scaling
- Cost optimization

PHASE 8: System Design (INTERVIEW LEVEL)

- Scalable backend systems
- AI system architecture
- Agent-based architectures
- Event-driven systems
- Microservices vs Monolith

FINAL PROJECTS (VERY IMPORTANT)

Must-Build Projects

1. Agentic AI SaaS (FastAPI + Agents + Pinecone)
2. AI Customer Support Agent
3. DevOps AI Assistant
4. Multi-agent Research System
5. AI Automation Platform